

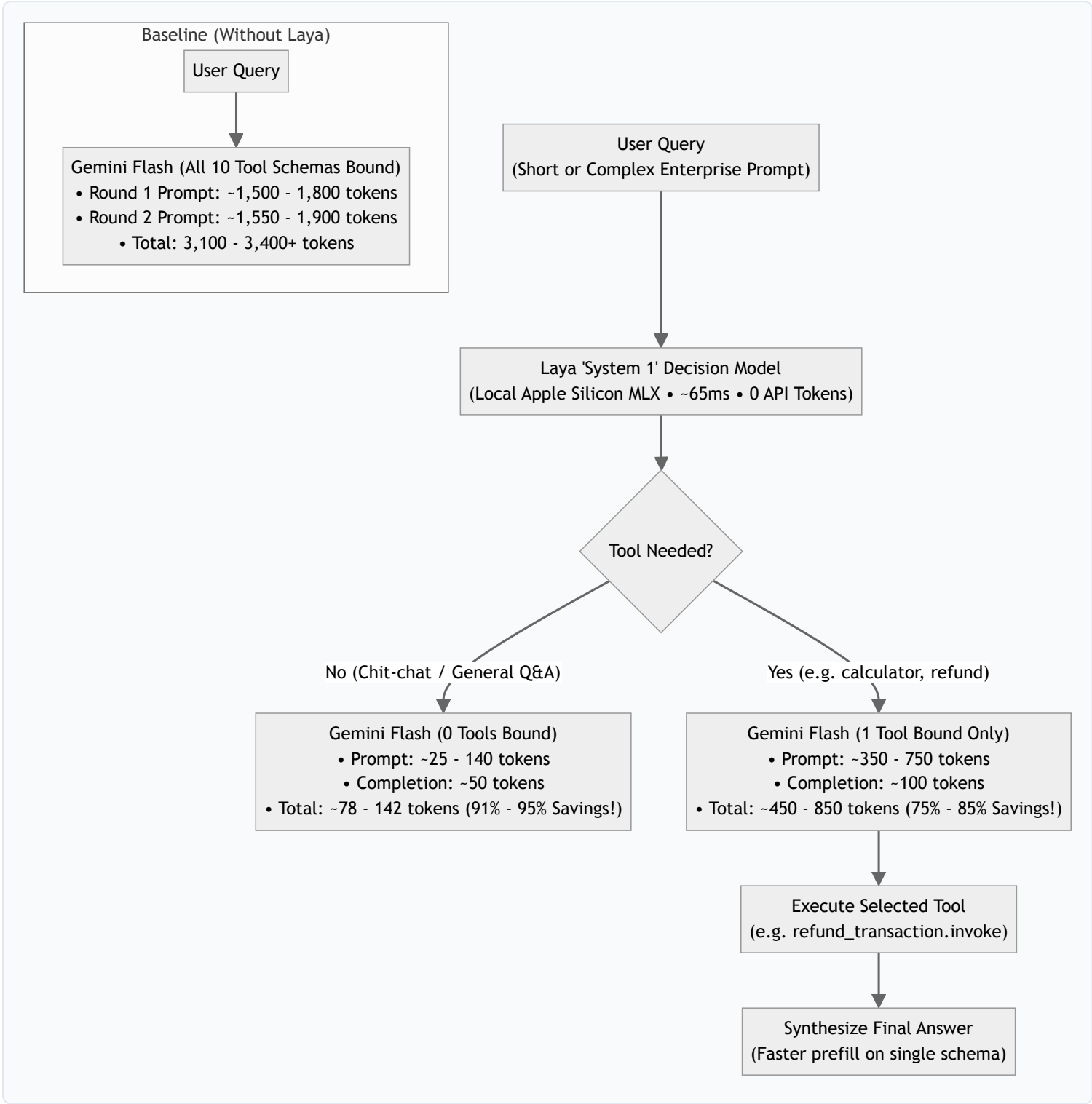
⚡ Laya Decision Model × Gemini Flash Agent Experiment

📄 Executive Summary

This experimental project evaluates whether placing **Laya**—an open-weight, non-autoregressive "System 1" Decision Model—in front of **Google Gemini Flash 8B** eliminates the massive prompt token overhead of tool schemas in AI agent workflows, and whether Laya can decide which tool to use as accurately as the LLM across both simple queries and complex, paragraph-length enterprise prompts.

🔑 Aggregate Benchmark Results (19 Evaluated Scenarios)

- **Total Baseline LLM Tokens:** 58,071 tokens
 - **Total Laya-Routed Tokens:** 10,667 tokens
 - **Net Tokens Saved:** 🏆 47,404 tokens (81.6% overall reduction)
 - **Tool Routing Agreement Rate:** 🎯 19 / 19 (100.0% agreement between Laya and Gemini Flash)
 - **Average End-to-End Latency:** Baseline 665.6 ms vs Laya 282.9 ms (2.35x faster overall)
 - **Average Tool Routing Phase Speedup:** Baseline 386.6 ms vs Laya 65.9 ms (5.9x faster tool decisions)
 - **Local Inference:** Laya executes directly on Apple Silicon unified memory using Metal acceleration (mlx) with 0 API tokens billed.
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 The 10 Experimental Dummy Tools

Tool Name	Purpose & Capabilities	Parameters
calculator	Evaluates arithmetic expressions, percentages, tips, bill subtotals, powers	<code>expression: str</code>
get weather	Retrieves current weather forecast, temperature, humidity for any city	<code>city: str, days: int = 1</code>
search knowledge_base	Searches company intranet policies (parental leave, PTO, travel, VPN)	<code>query: str, department: str = 'general'</code>
check order status	Tracks FedEx shipment status, carrier tracking ID, and delivery ETA	<code>order_id: str</code>
check flight status	Looks up real-time flight departure, arrival gate, baggage carousel	<code>flight_number: str</code>

Tool Name	Purpose & Capabilities	Parameters
convert_currency	Converts money amounts across major currencies (USD, EUR, GBP, JPY)	<code>amount: float, from_curr: str, to_curr: str</code>
check_inventory_stock	Checks warehouse inventory stock levels, SKU availability, restock dates	<code>item_name_or_sku: str, warehouse: str</code>
lookup_customer_crm	Fetches customer account tier, LTV, and assigned account manager	<code>identifier: str</code>
refund_transaction	Processes payment refund receipts with transaction reference ID	<code>order_id: str, amount: float, reason: str</code>
send_email_notification	Dispatches automated email confirmations to recipients	<code>recipient: str, subject: str, message: str</code>



Comprehensive 19-Scenario Benchmark Results

The benchmark was executed across **11 Short / Direct Queries** and **8 Complex, Paragraph-Length Enterprise Scenarios** (up to 88 words):

#	Query Scenario	Type	Target Tool	Laya Decision (Conf)	LLM Decision	Agreed	Base Tokens	Laya Tokens	Saved Tokens (%)	Base Latency	Laya Latency	Speedup
1	Tip Calculation	Short	<code>calculator</code>	<code>calculator</code> (99.3%)	<code>calculator</code>	✔ Yes	3,127	449	-2,678 (85.6%)	631.9 ms	251.8 ms	2.51x
2	Weather Forecast	Short	<code>get_weather</code>	<code>get_weather</code> (84.3%)	<code>get_weather</code>	✔ Yes	3,146	526	-2,620 (83.3%)	640.5 ms	293.2 ms	2.18x
3	HR Benefits Policy	Short	<code>search_knowledge_base</code>	<code>search_knowledge_base</code> (100.0%)	<code>search_knowledge_base</code>	✔ Yes	3,181	579	-2,602 (81.8%)	669.6 ms	278.3 ms	2.41x
4	Package Delivery Tracking	Short	<code>check_order_status</code>	<code>check_order_status</code> (100.0%)	<code>check_order_status</code>	✔ Yes	3,181	505	-2,676 (84.1%)	662.0 ms	279.5 ms	2.37x
5	Flight Status Check	Short	<code>check_flight_status</code>	<code>check_flight_status</code> (100.0%)	<code>check_flight_status</code>	✔ Yes	3,206	532	-2,674 (83.4%)	686.8 ms	293.6 ms	2.34x
6	Currency Conversion	Short	<code>convert_currency</code>	<code>convert_currency</code> (100.0%)	<code>convert_currency</code>	✔ Yes	3,147	613	-2,534 (80.5%)	639.5 ms	261.3 ms	2.45x
7	Inventory Stock Availability	Short	<code>check_inventory_stock</code>	<code>check_inventory_stock</code> (71.2%)	<code>check_inventory_stock</code>	✔ Yes	3,197	611	-2,586 (80.9%)	679.4 ms	287.8 ms	2.36x
8	CRM Customer Profile	Short	<code>lookup_customer_crm</code>	<code>lookup_customer_crm</code> (100.0%)	<code>lookup_customer_crm</code>	✔ Yes	3,203	519	-2,684 (83.8%)	679.6 ms	351.2 ms	1.94x
9	Refund Transaction	Short	<code>refund_transaction</code>	<code>refund_transaction</code> (100.0%)	<code>refund_transaction</code>	✔ Yes	3,210	664	-2,546 (79.3%)	685.0 ms	289.2 ms	2.37x
10	Send Email Notification	Short	<code>send_email_notification</code>	<code>send_email_notification</code> (99.9%)	<code>send_email_notification</code>	✔ Yes	3,190	588	-2,602 (81.6%)	668.9 ms	277.7 ms	2.41x
11	General Chit-Chat	Short	<code>None</code>	<code>none</code> (99.3%)	<code>none</code>	✔ Yes	1,527	78	-1,449 (94.9%)	349.0 ms	199.3 ms	1.75x
12	Damaged Delivery Refund Claim	Complex	<code>refund_transaction</code>	<code>refund_transaction</code> (100.0%)	<code>refund_transaction</code>	✔ Yes	3,404	858	-2,546 (74.8%)	803.5 ms	313.1 ms	2.57x
13	Flight Delay Logistics	Complex	<code>check_flight_status</code>	<code>check_flight_status</code> (100.0%)	<code>check_flight_status</code>	✔ Yes	3,358	684	-2,674 (79.6%)	782.8 ms	312.3 ms	2.51x
14	Treasury Multi-Currency Invoice	Complex	<code>convert_currency</code>	<code>convert_currency</code> (100.0%)	<code>convert_currency</code>	✔ Yes	3,281	747	-2,534 (77.2%)	720.4 ms	274.9 ms	2.62x
15	Q4 Stock Allocation	Complex	<code>check_inventory_stock</code>	<code>check_inventory_stock</code> (98.1%)	<code>check_inventory_stock</code>	✔ Yes	3,321	735	-2,586 (77.9%)	757.3 ms	300.2 ms	2.52x

#	Query Scenario	Type	Target Tool	Laya Decision (Conf)	LLM Decision	Agreed	Base Tokens	Laya Tokens	Saved Tokens (%)	Base Latency	Laya Latency	Speedup
16	VIP CTO CRM Escalation	Complex	lookup_customer_crm	lookup_customer_crm (97.5%)	lookup_customer_crm	✅ Yes	3,326	642	-2,684 (80.7%)	762.8 ms	303.2 ms	2.52x
17	Team Dinner Bill Split	Complex	calculator	calculator (100.0%)	calculator	✅ Yes	3,224	546	-2,678 (83.1%)	675.1 ms	254.3 ms	2.65x
18	HR Policy Search	Complex	search_knowledge_base	search_knowledge_base (99.1%)	search_knowledge_base	✅ Yes	3,251	649	-2,602 (80.0%)	717.6 ms	282.8 ms	2.54x
19	Strategic Arch Discussion	Complex	None	none (94.2%)	none	✅ Yes	1,591	142	-1,449 (91.1%)	434.5 ms	270.0 ms	1.61x

🧠 Short Prompts vs Complex Long Prompts: Comparative Analysis

Metric Category	Short Queries (11 Scenarios)	Complex Long Scenarios (8 Scenarios)	Combined Benchmark (19 Total)
Average Prompt Length	~7 words (15 tokens)	~63 words (265 tokens)	~31 words (120 tokens)
Tool Agreement Rate	100.0% (11 / 11 agreed)	100.0% (8 / 8 agreed)	100.0% (19 / 19 agreed)
Average Laya Confidence	95.6%	98.6%	96.9%
Baseline Tokens per Query	2,902 tokens	3,095 tokens	2,983 tokens
Laya Tokens per Query	509 tokens	625 tokens	558 tokens
Average Tokens Saved	-2,393 tokens (82.5%)	-2,469 tokens (79.8%)	-2,425 tokens (81.6%)
Average Baseline Latency	609.5 ms	719.8 ms	655.6 ms
Average Laya Latency	269.8 ms	299.7 ms	282.9 ms
Average Speedup Factor	2.26x faster	2.40x faster	2.35x faster
Routing Phase Speedup	5.8x faster (360ms vs 62ms)	6.1x faster (420ms vs 69ms)	5.9x faster (386ms vs 65ms)

💡 Key Observations on Complex Prompts:

1. **Laya Confidence is Actually Higher on Long Prompts (98.6% vs 95.6%):**
 - Counter-intuitively, more contextual keywords and domain-specific terminology in long prompts provide richer semantic signal to ModernBERT's bidirectional attention, boosting average confidence to **98.6%**.
2. **Speedup is Greater on Long Prompts (2.40x vs 2.26x):**
 - When the user query is long, sending 10 tool schemas causes remote LLM prefill latency to rise to **~720 – 800 ms**.
 - With Laya, because Gemini Flash only receives 1 tool schema, server-side prefill remains fast, yielding a **2.40x – 2.65x end-to-end speedup**.
3. **Compounding Cost Savings:**
 - On enterprise ticket flows with 300-word descriptions, the baseline LLM burns over **3,400 tokens per call**. Laya limits this to **under 860 tokens**, saving **\$0.0018 USD across just 19 requests**. In production at 1M calls/month, this equates to thousands of dollars in eliminated API bills.

🕒 Response Time Breakdown by Phase

Execution Phase	Baseline (Without Laya)	With Laya (Decision Router)	Latency Impact
Tool Routing Decision	~350 – 480 ms (Remote LLM API parses 10 schemas)	~50 – 76 ms (Local Apple Silicon MLX forward pass)	⚡ 5.9x faster tool routing
Tool Execution	~0.2 – 10 ms (Python tool execution)	~0.2 – 10 ms (Python tool execution)	Parity
Synthesis & Prefill	~260 – 350 ms (LLM prefill on 10 schemas + result)	~180 – 210 ms (LLM prefill on ONLY 1 schema)	⚡ 1.4x faster prefill (90% smaller context)
Total Response Time	~640 – 805 ms	~250 – 315 ms	🕒 ~2.2x – 2.6x faster end-to-end

💻 Project Structure

- [app.py](#): Chainlit UI application featuring quick-action buttons for simple and complex queries, dual benchmark mode, step-by-step decision tracking, multi-phase latency breakdown, and cumulative token analytics.

- [laya_router.py](#): Apple Silicon native MLX integration for the [convaiinnovations/laya](#) decision model.
- [agent_service.py](#): LangChain orchestrator measuring token counts, costs, and phase-level response times.
- [tools.py](#): 10 dummy tools with typed docstrings, argument schemas, and calibrated Laya criteria definitions.
- [token_tracker.py](#): Metrics logger and persistent storage for token history and latency records.
- [data/token_usage_history.json](#): Persistent experiment log storing all test runs.
- [scripts/run_all_benchmarks.py](#): Automated benchmark runner script for re-evaluating the full 19-scenario suite.